Midterm Examination
Numerical Analysis

June 12, 2025

Name:	
Student ID:	

Important Notes:

- **Do NOT open this booklet until the start signal is given.** Opening it before then will be considered cheating.
- Submit all answers through Moodle using Safe Exam Browser (SEB) and the provided seb file.
- In case of system issues, record your selected answers in this booklet.
- If there is a discrepancy between your Moodle answers and notes in this booklet, the Moodle answers will take precedence.
- **After confirming that your submission through Moodle has been completed successfully**, take this booklet with you after the examination.

Q. 1-1

Consider a floating-point system with $\beta = 2$, $p = 4$, and $-2 \leq e \leq 3$. Determine the correctness of the following statements.

a. The largest number in this system is 15.

b. The smallest positive normalized number in this system is 0.0625.

c. There exists a gap between 1.5 and 1.75 where no numbers can be represented exactly in this system.

Q. 1-2

Determine the correctness of the following statements about `binary64` floating-point representation.

a. In binary64, the number 0.1 (decimal) can be represented exactly.

b. When underflow occurs in binary64, the result is always set to zero.

c. In the denormalized region, all representable numbers have the same absolute spacing between consecutive values.

Q. 1-3

Consider the expressions $(1+\eta)-1$ and $1+(\eta-1)$ for a small number η in binary64 floating-point arithmetic. For $\eta = 2^{-52}, 2^{-53}, 2^{-54}$, select the correct answer from the following options:
Hint: You can consider $1-(1-\eta)$ in place of $1+(\eta-1)$.

a. The two expressions always give the same result for any $\eta = 2^{-52}, 2^{-53}, 2^{-54}$.

b. The two expressions always give different results for any $\eta = 2^{-52}, 2^{-53}, 2^{-54}$.

c. The expressions give different results when $\eta = 2^{-52}, 2^{-53}$, while the other give the same result.

d. The expressions give different results when $\eta = 2^{-53}, 2^{-54}$, while the other give the same result.

e. The expressions give different results when $\eta = 2^{-54}, 2^{-52}$, while the other give the same result.

f. The expressions give different results when $\eta = 2^{-54}$, while the other two give the same result.

g. The expressions give different results when $\eta = 2^{-53}$, while the other two give the same result.

h. The expressions give different results when $\eta = 2^{-52}$, while the other two give the same result.

Q. 1-4

A measurement device has a specified accuracy such that the absolute error should not exceed 0.05 units. If the true value is 10.0 units, what is the maximum allowable relative error?

Q. 1-5

Let $\hat{x}$ and $\hat{y}$ be approximations to x and y with relative errors ϵ_x and ϵ_y respectively. Determine the correctness of the following statements about error propagation.

a. The relative error in $\hat{x} \cdot \hat{y}$ is approximately $\epsilon_x + \epsilon_y$ for small errors.

b. The absolute error in $\hat{x} + \hat{y}$ is approximately $|x|\epsilon_x + |y|\epsilon_y$ for small errors.

c. The relative error in $\hat{x}/\hat{y}$ is approximately $\epsilon_x - \epsilon_y$ for small errors.

Q. 1-6

Determine the correctness of the following statements about floating-point numbers.

a. The sum of two binary64 numbers is always exactly representable in binary64.

b. In binary64 arithmetic, numerical algorithms can achieve a relative error of about 8 decimal digits in the best case.

c. Cancellation is primarily an issue in subtraction; it does not affect addition.

d. The expression $\sqrt{x+1} - \sqrt{x}$ becomes numerically unstable for large x.

e. Computing $e^x - 1$ for small $x > 0$ is numerically stable.

Q. 1-7

For the quadratic equation $x^2 - 2px + q = 0$ where $p \gg \sqrt{q}$, which formulation gives the most accurate smaller root?

a. $x = p - \sqrt{p^2 - q}$

b. $x = \frac{q}{p+\sqrt{p^2-q}}$

c. $x = \frac{-(-2p)-\sqrt{4p^2-4q}}{2}$

d. All formulations are equally accurate

Q. 2-1

For the vector $x = \begin{pmatrix} 3 \\ -4 \\ 0 \\ 12 \end{pmatrix}$, what is $\|x\|_1$, $\|x\|_2$, and $\|x\|_\infty$?

Q. 2-2

Determine the correctness of the following statements:

a. The triangle inequality property of a norm is stated as $\|x + y\| \leq \|x\| + \|y\|$.

b. The operator norm of a matrix A is defined as $\|A\| = \min_{\|x\|=1} \|Ax\|$.

c. The Frobenius norm of a matrix is always greater than or equal to its spectral norm (ℓ_2 norm).

d. For any matrix A, the equality $\|A\|_1 = \|A^T\|_\infty$ holds.

e. The condition number of a matrix is independent of the norm used.

f. For any invertible matrix A, $\text{cond}(A^{-1}) = \text{cond}(A)$.

g. Matrix $A = \begin{pmatrix} 5 & 1 & -1 \\ 2 & 8 & 2 \\ -1 & 1 & 6 \end{pmatrix}$ is strictly diagonally dominant.

Q. 2-3

Determine the correctness of the following statements about iterative methods for solving $Ax = b$. Let M be the iteration matrix.

a. The Gauss-Seidel method is guaranteed to converge if the Jacobi method converges.

b. A sufficient condition for the convergence of both Jacobi and Gauss-Seidel methods is that the matrix A is symmetric positive-definite.

c. If a matrix A is strictly diagonally dominant, the Gauss-Seidel method is guaranteed to converge.

d. The Conjugate Gradient (CG) method can be applied to any non-singular matrix A without preconditioning.

Q. 2-4

Consider solving the linear system $Ax = b$. The condition number of matrix A is $\text{cond}_\infty(A) = 5000$. The relative error in the vector b is measured in the infinity norm as $\frac{\|\Delta b\|_\infty}{\|b\|_\infty} = 0.001$. Estimate the maximum possible relative error in the solution, $\frac{\|\Delta x\|_\infty}{\|x\|_\infty}$. Write your answer as an integer or decimal in the answer box.

Q. 2-5

Consider the stationary iterative method $x_{k+1} = Mx_k + c$, where M is the iteration matrix and c is a constant vector. Three iterative methods for solving a linear system produce iteration matrices $M = M_A, M_B, M_C$ with the following spectral radii: $\rho(M_A) = 0.5$, $\rho(M_B) = 1.2$, and $\rho(M_C) = 0.9$. Which method will be expected to converge the fastest?

a. Method A

b. Method B

c. Method C

d. All methods converge at the same rate.

Q. 3-1

Determine the correctness of the following statements:

a. Newton's method to solve $f(x) = 0$ is defined by the iterative formula $x_{k+1} = x_k - \frac{f'(x_k)}{f(x_k)}$, where $f'(x_k)$ is the first derivative of f at x_k.

b. Newton's method exhibits quadratic convergence under a suitable condition.

c. Newton's method is guaranteed to converge regardless of the initial guess.

d. For a system of nonlinear equations, Newton's method requires solving a system of linear equations with the Hessian matrix.

e. The Simplified Newton's method has a faster convergence rate than the standard Newton's method.

f. The Durand-Kerner method is used for finding a single real root of a polynomial with high accuracy.

Q. 3-2

To solve the equation $f(x) = x^2 - 3 = 0$ (whose positive root is $x^* = \sqrt{3}$), consider the following five fixed-point iteration schemes. Which scheme exhibits the fastest convergence rate near the positive root?

a. $g_1(x) = \frac{3}{x}$

b. $g_2(x) = \frac{x}{2} + \frac{3}{2x}$

c. $g_3(x) = x^2 - x + 3$

d. $g_4(x) = x - \frac{1}{4}(x^2 - 3)$

e. $g_5(x) = -x^2 + x + 3$

f. All schemes have the same convergence rate.

Q.4

Instructions: Below are pairs of prompts (instructions) given to a generative AI and the Python code output in response. Evaluate the correctness of each code.

1. If there is an error, enter the line number (maximum one) in the answer field as an integer value.
2. If there are multiple errors, you only need to input one of them.
3. If there are no errors, enter 0.

- Language: Assumes the latest Python 3
- Environment: All necessary packages (NumPy, etc.) are assumed to be installed and imported
- Note: Simple syntax errors as a programming language are not expected. Evaluate logical errors in algorithms and mathematical formulas.

Q. 4-1

Prompt

Please implement a numerically stable version of the function $f(x) = \sqrt{x+1} - 1$ that avoids cancellation when x is very close to 0.

Output

```
1     def calculate_stable(x):
2         # Use a more stable expression
3         return np.sqrt(x) * (np.sqrt(1 + 1/x) - 1/np.sqrt(x))
4
5     # Test value for x
6     x = 1e-16
7     # Calculate
8     result = calculate_stable(x)
9     # Output calculation result
10    print(f"For x={x}, the calculation result is {result}.")
```

Q. 4-2

Prompt

Create a program to solve the nonlinear equation $\sin(x) - 0.5x = 0$ using a fixed-point iterative method.

Code

```python
def g(x):
    # Function g(x) for the given fixed-point iteration
    return np.sin(x) - 0.5*x

def fixed_point_iteration(g_func, x0, max_iter=30, tol=1e-8):
    x = x0
    print(f"k=0, x = {x:.8f}")

    for k in range(max_iter):
        x_new = g_func(x)
        print(f"k={k+1}, x = {x_new:.8f}")

        # Convergence check
        if np.abs(x_new - x) < tol:
            print(f"\nConverged.")
            return x_new

        x = x_new

    print("\nReached maximum number of iterations.")
    return x

# --- Execution ---
# Set initial value
initial_guess = 1.0
print(f"Initial value x0 = {initial_guess}")

# Execute fixed-point iteration
solution = fixed_point_iteration(g, initial_guess)
print(f"Final value: {solution:.8f}")
```

Q. 4-3

Prompt

Create a Python code to find the solution for the following system of nonlinear equations using the 2D Newton's method. $f_1(x, y) = x^2 + y - 3 = 0$, $f_2(x, y) = x + y^2 - 5 = 0$

Code

```python
def F(x):
    return np.array([x[0]**2 + x[1] - 3,
                     x[0] + x[1]**2 - 5])

def J(x):
    # Jacobian of F(x)
    return np.array([[2 * x[0], 1],
                     [1, 2 * x[1]]])

def newton_method_2d(x0, max_iter=20, tol=1e-8):
    x = np.array(x0, dtype=float)
    # Calculate Jacobian
    Jx = J(x)

    for k in range(max_iter):
        Fx = F(x)
        if np.linalg.norm(Fx) < tol:
            print(f"Solution converged to {x}.")
            return x

        # Calculate the update vector
        x = x - np.linalg.solve(Jx, Fx)

    print("Reached maximum number of iterations.")
    return x

# Calculate from initial value [2.0, 2.0]
solution = newton_method_2d([2.0, 2.0])
```